HOLLOW DESIGN ACCEPTED

MBB: summary

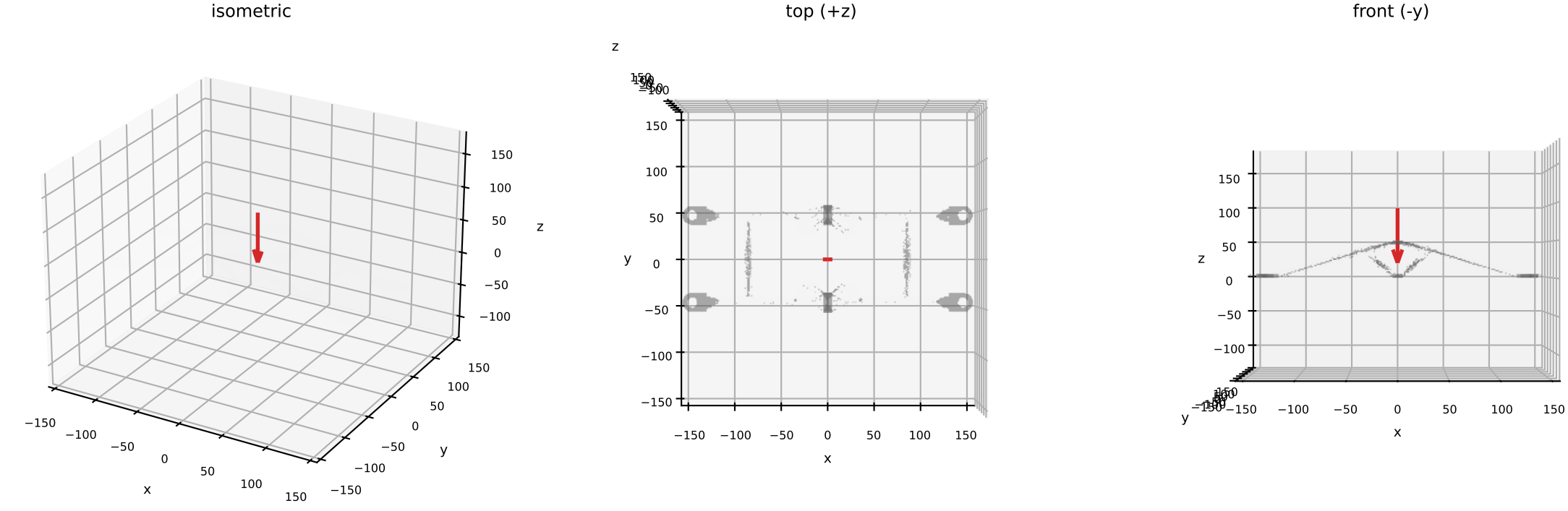
```
cshell design accepted: J = +0.4812 > ln(1 + 0.02) with K/K0 ≥ 1.096, P/P0 ≥ 2.390 at every resolution [body-
fitted check not run: model exceeds solver capacity; voxel criteria only]

Mass check: {'target': 88.6399839761109, 'final': 88.64753922782675, 'error': 8.523525588488567e-05, 'V0_mm3': 85230.75382318355, 'density_g_mm3': 0.0010400000000000001, 'mass_matched': True}

Comparison (solid, uniform shell, vshell best, cshell starts, export)
        label           K_K0           P_P0              J      governing          w_min          w_max
        solid              1              1              0       buckling
uniform shell          0.775          1.109       -0.07571       strength          2.214          2.214
  vshell best              1              1        0.00035       strength          1.565          1.765
cshell unifor          1.097          2.493         0.5031                         1.502          5.529
cshell vshell          1.098          2.492         0.5032                         1.502          5.529
cshell neutra          1.098          2.493         0.5032                         1.502          5.527
cshell export          1.097          2.391          0.482       buckling        0.02378          5.527

Run settings / other summary keys
  schema: cshell.summary/1
  outcome: cshell design accepted
  criteria: {'g_min': 0.02, 'ln_1_g_min': 0.01980262729617973, 'f_K': 0.85, 'f_P': 0.85}
  best: {'start': 'neutral', 'J_opt': 0.5032214773938207, 'D_comp': 8.0, 'xi_file': 'xi_neutral_phase1.npy', 'phase': 1}
  starts: [{'start': 'uniform', 'status': 'done', 'feasible': True, 'phases_done': 2, 'iterations': 98, 'J_opt':
0.5030762573491533, 'K_K0_opt': 1.097035255673686, 'P_P0_opt': 2.4931356007447976, 'mass_err': -1.7871120070789 28e-05,
'hours': 0.5544665444559521, 'D_comp': 8.0, 'xi_file': 'xi_uniform_phase1.npy', 'w_min': 1.5024765334132488, 'w_max':
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'seconds': 28.072948694229126, 'error': None, 'pcg': [[7, 8, 0.00022535301898698784], [6, 6, 0.00022535301898698784],
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```

MBB: original geometry; loads [[0, 0, -200]] N (arrow, indicative placement); supports ['cylinder'] (face selection from config only; arrays.npz absent)



MBB: design space on Gamma_h

r_MA [mm] (isometric)

r_MA [mm] (top (+z))

r_MA [mm] (front (-y))

d_hi [mm] (isometric)

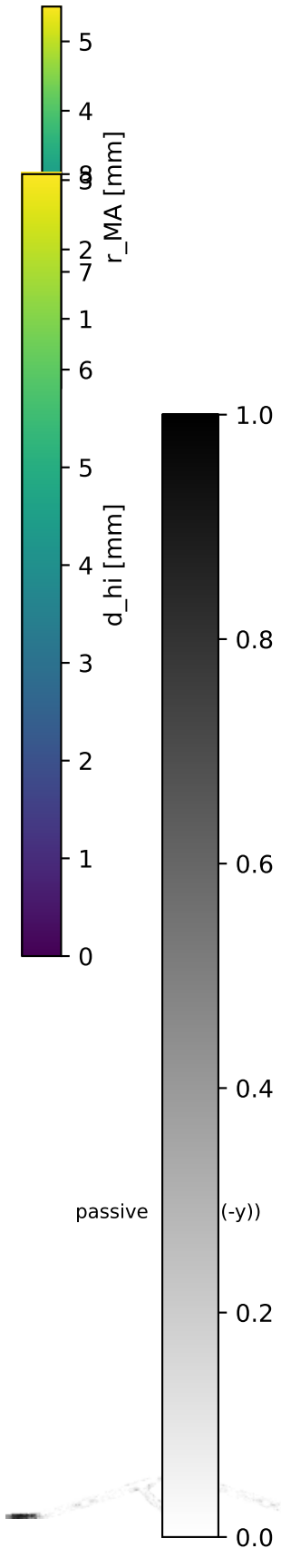
d_hi [mm] (top (+z))

d_hi [mm] (front (-y))

passive (isometric)

passive (top (+z))

passive (front (-y))

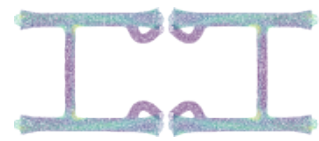


MBB: result fields

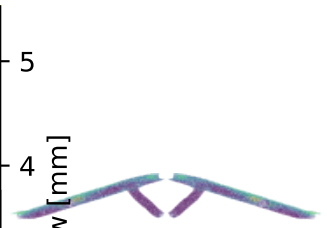
wall w [mm] (isometric)



wall w [mm] (top (+z))



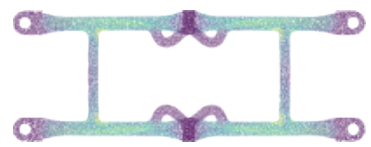
wall w [mm] (front (-y))



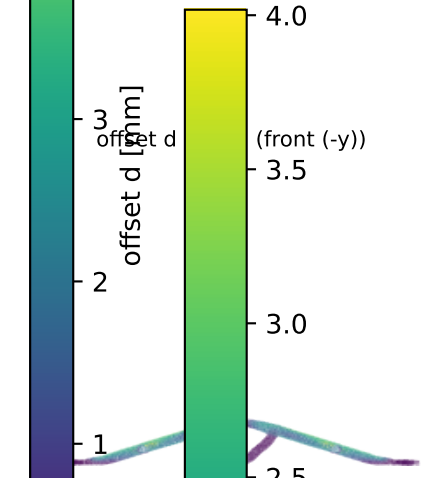
offset d [mm] (isometric)



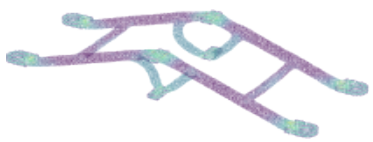
offset d [mm] (top (+z))



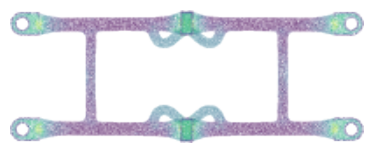
offset d [mm] (front (-y))



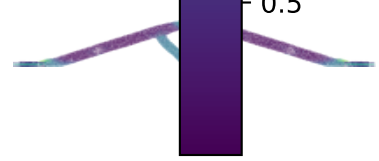
inset a [mm] (isometric)



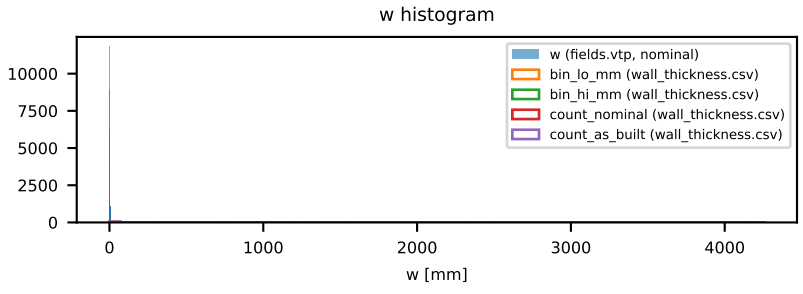
inset a [mm] (top (+z))



inset a [mm] (front (-y))

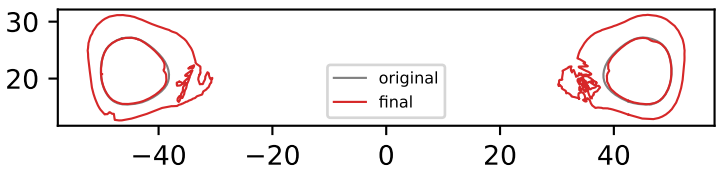


w histogram

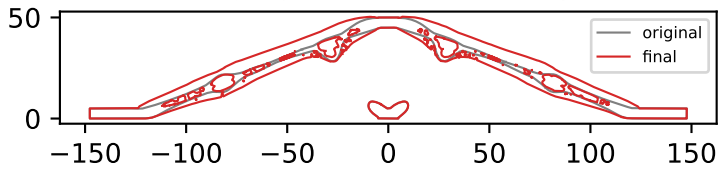


sections: original (grey) vs final (red)

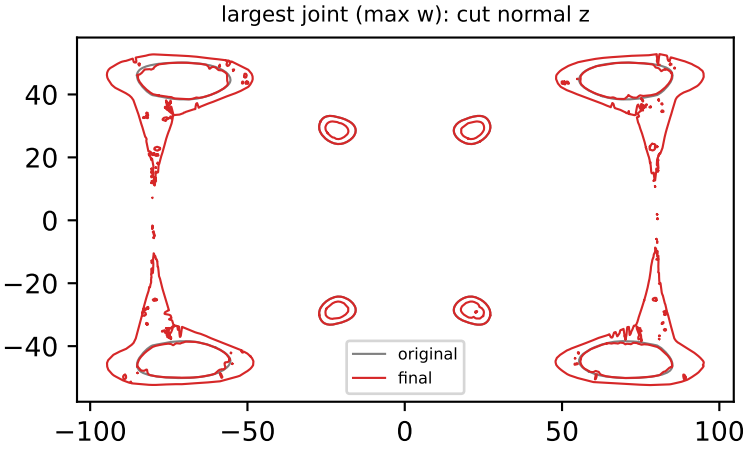
largest joint (max w): cut normal x



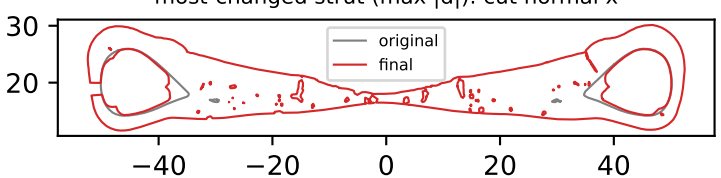
largest joint (max w): cut normal y



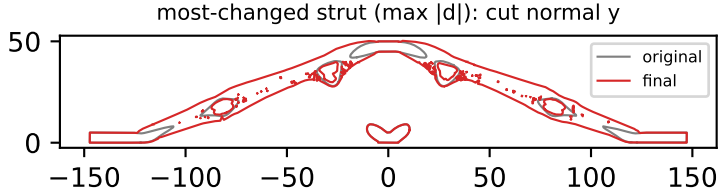
largest joint (max w): cut normal z



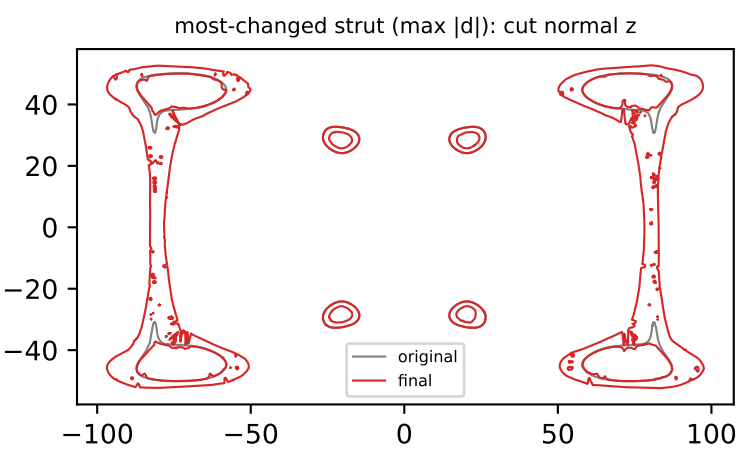
most-changed strut (max |d|): cut normal x



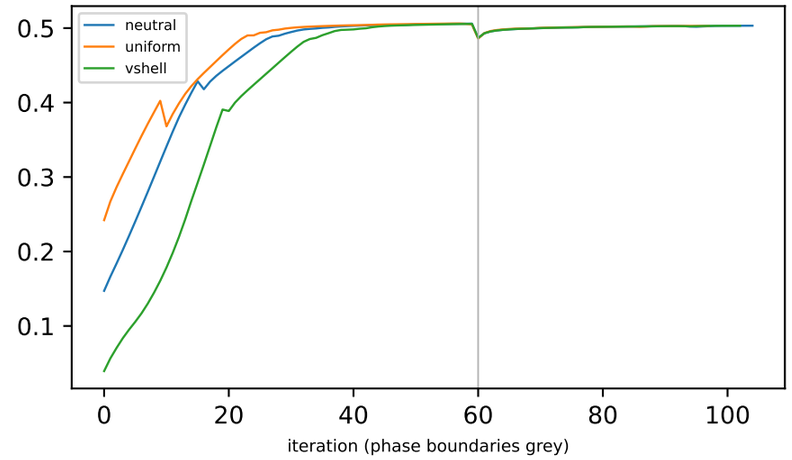
most-changed strut (max |d|): cut normal y



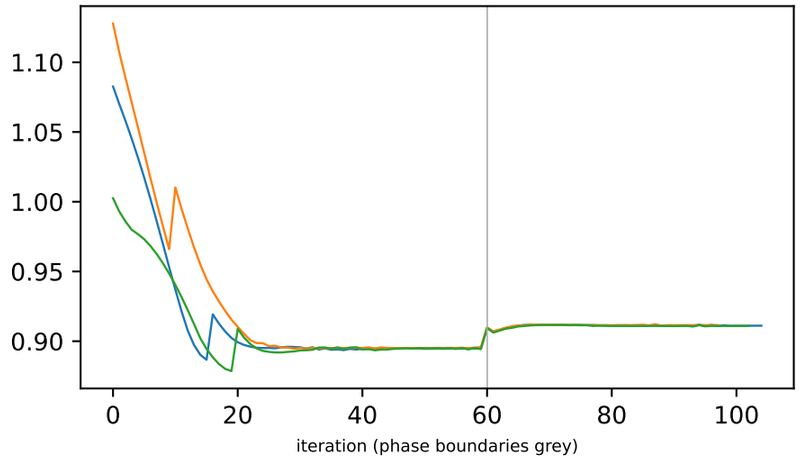
most-changed strut (max |d|): cut normal z



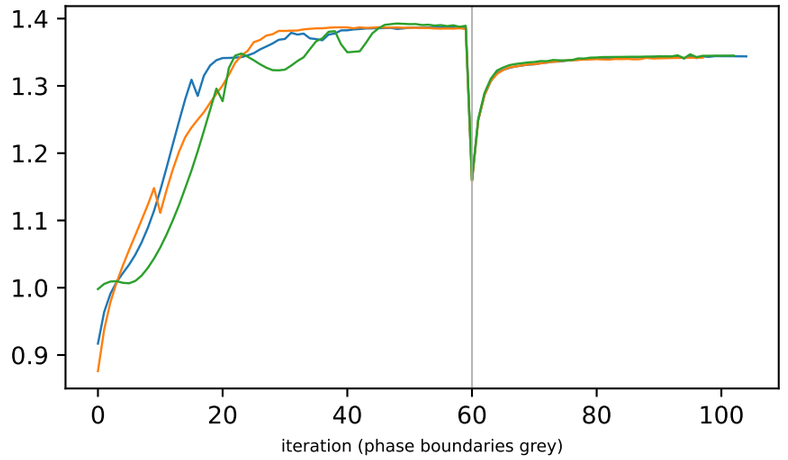
J



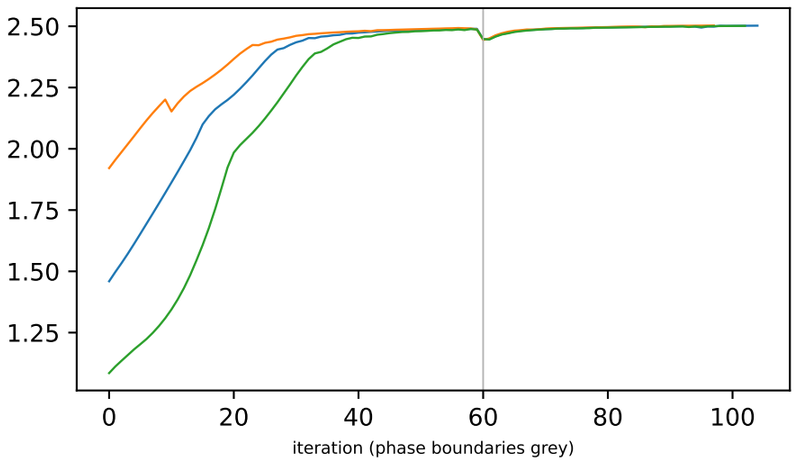
C/C0



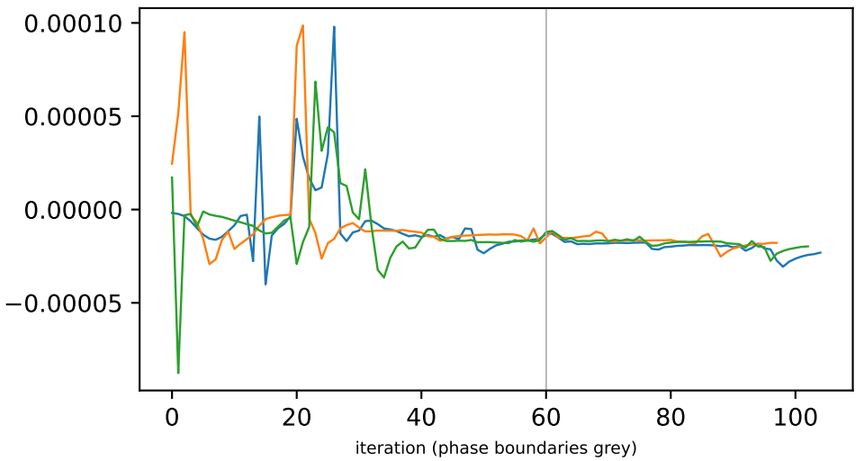
lam_s/lam_s0



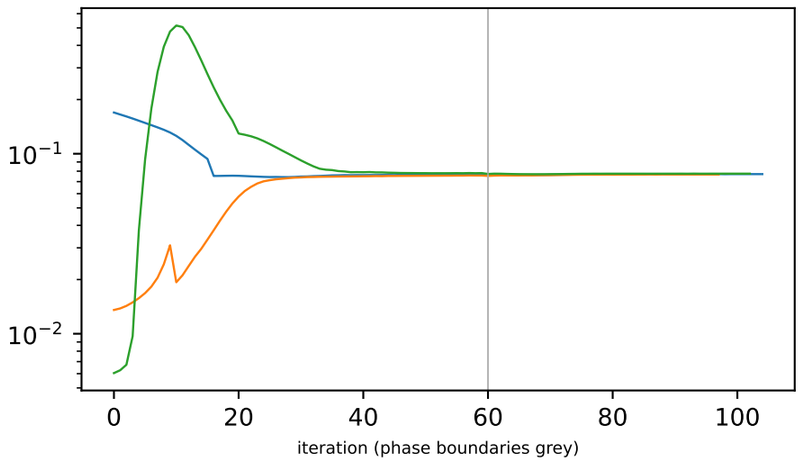
lam_b/lam_b0



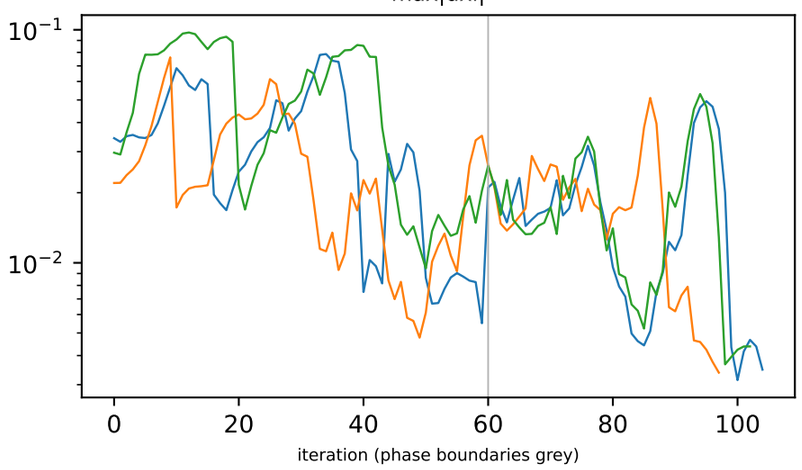
V/V0 - 1



G_s

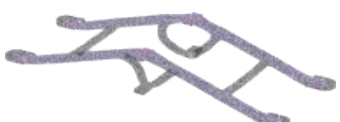


max|dxi|



MBB: material von Mises on final design (solid-baseline stress not stored in the run dir)

von Mises [MPa] (isometric)

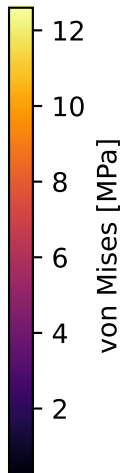


von Mises [MPa] (top (+z))



von Mises [MPa] (front (-y))

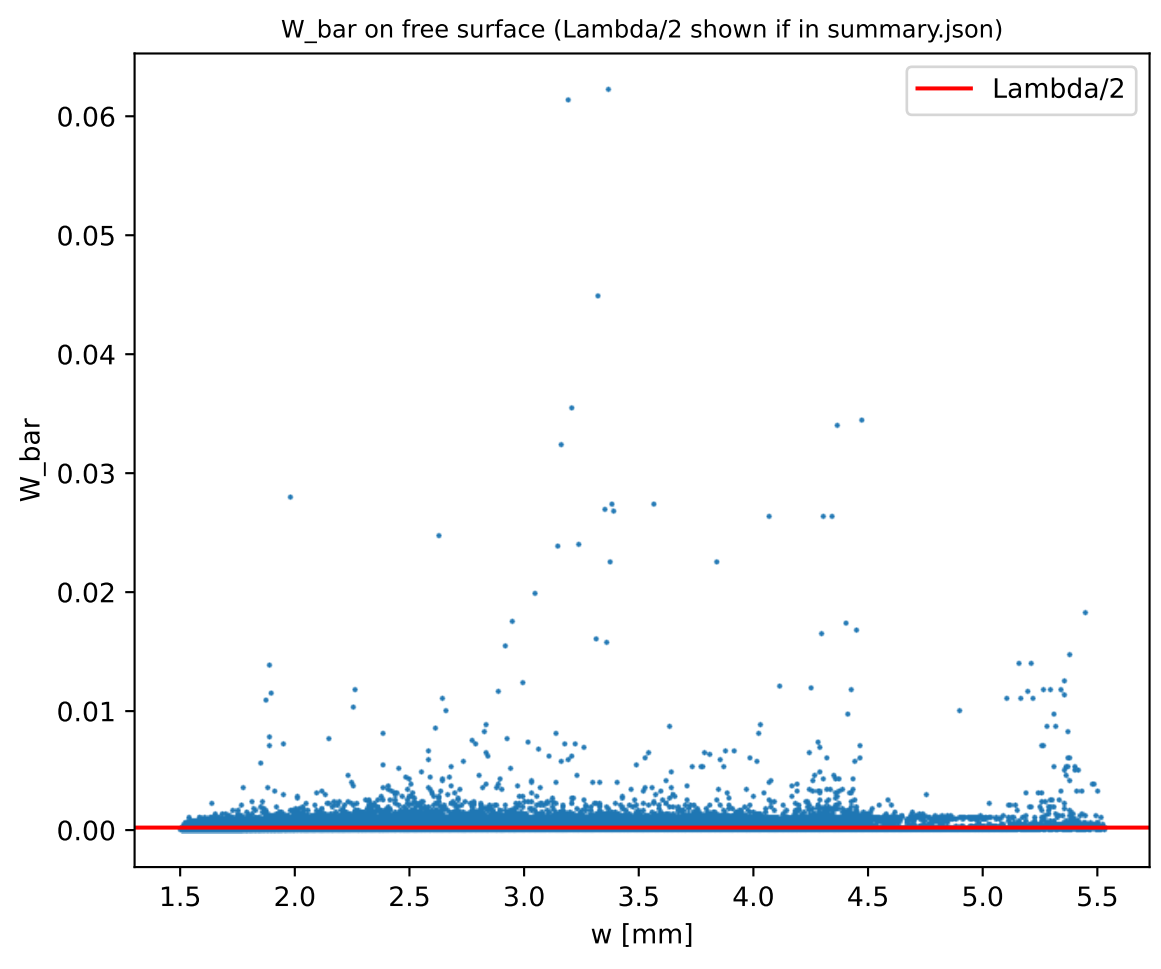


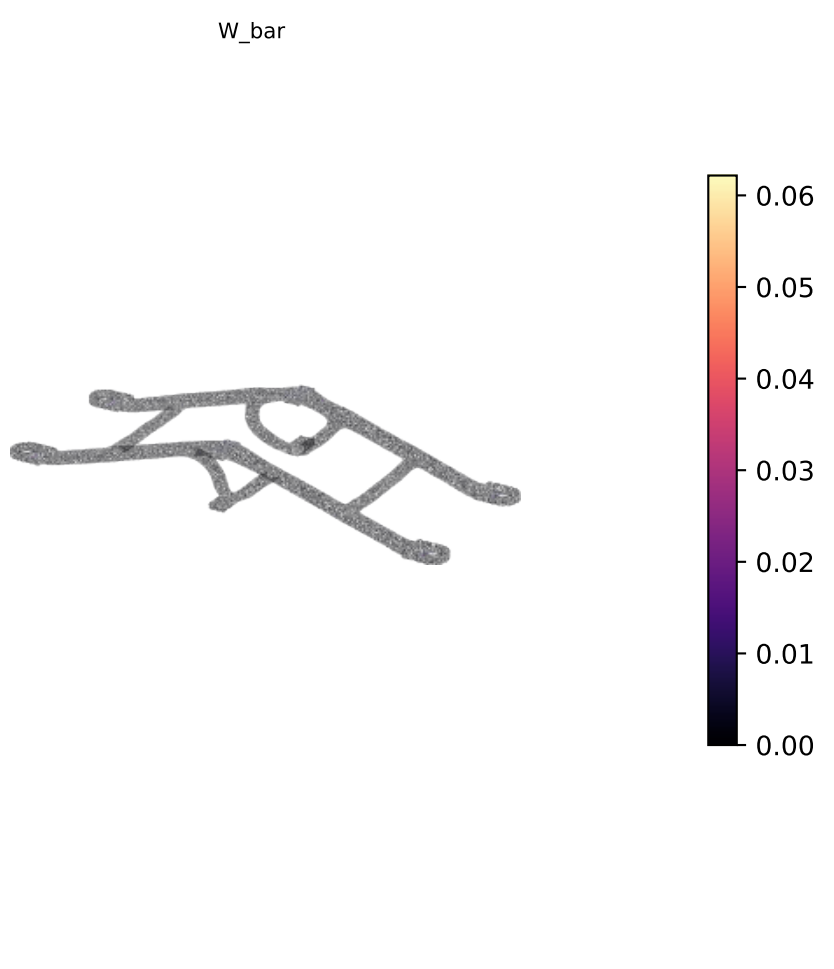


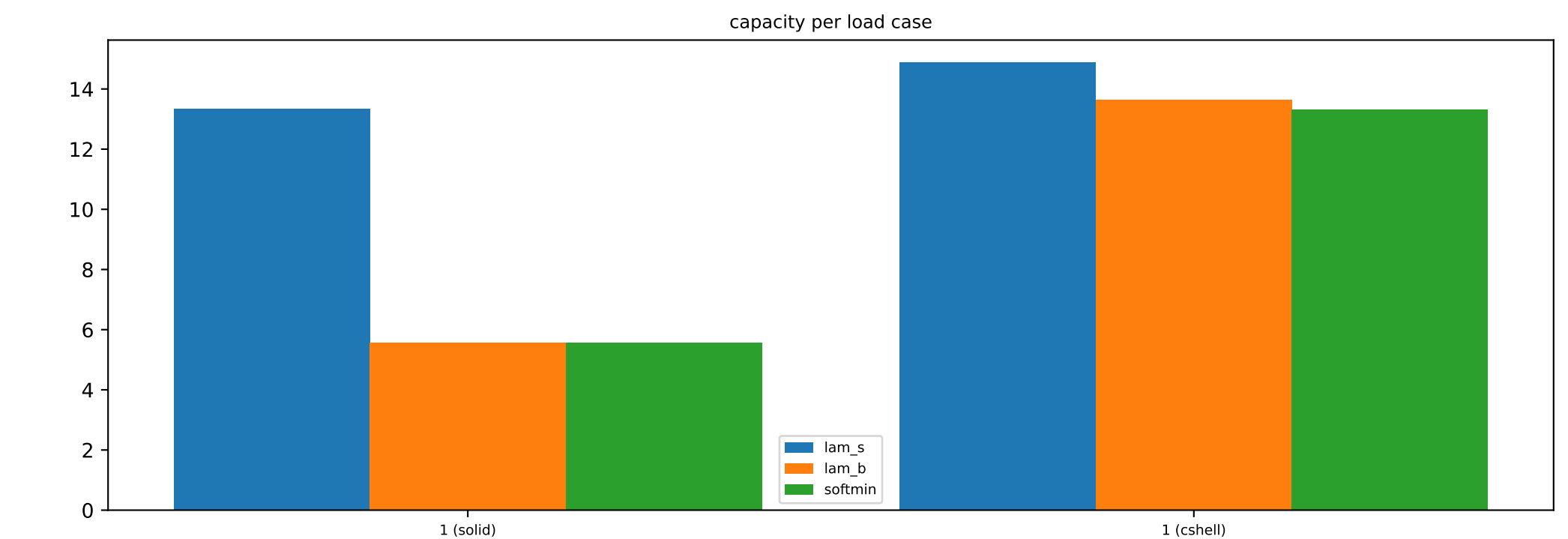
08_buckling: buckling

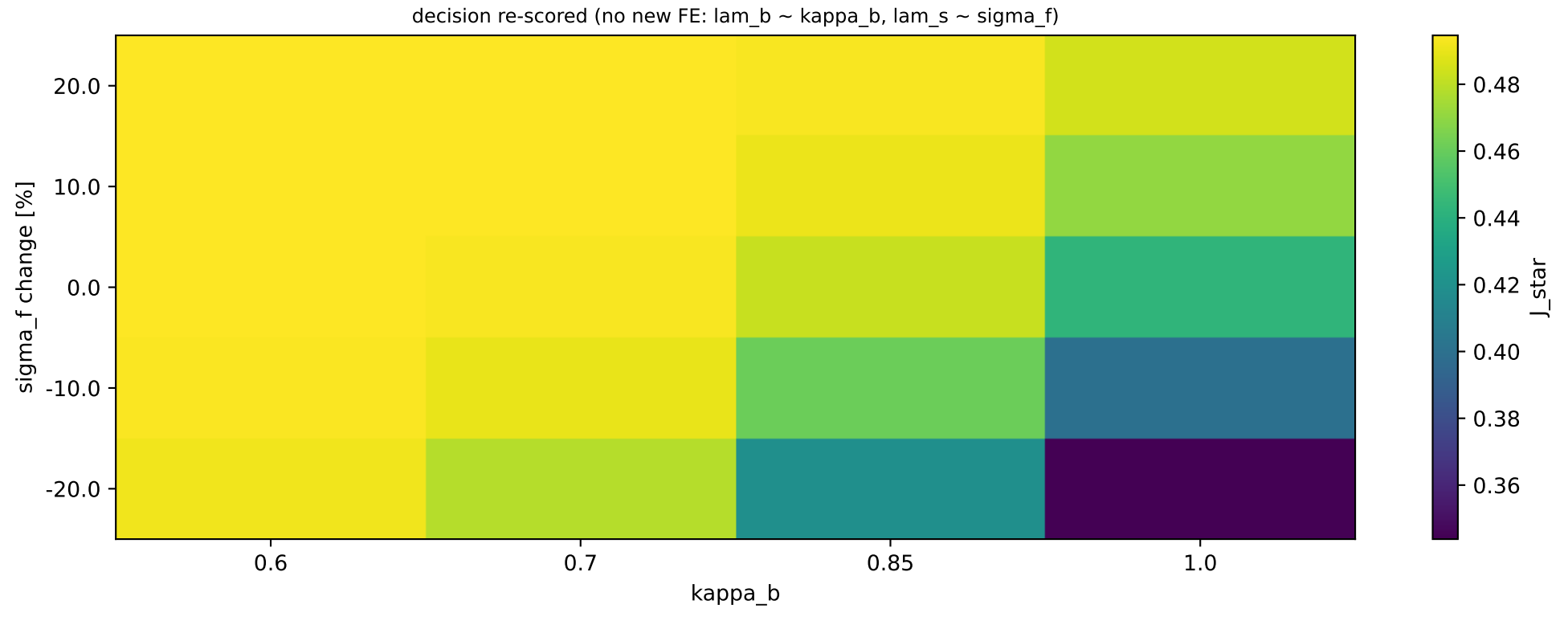
not available

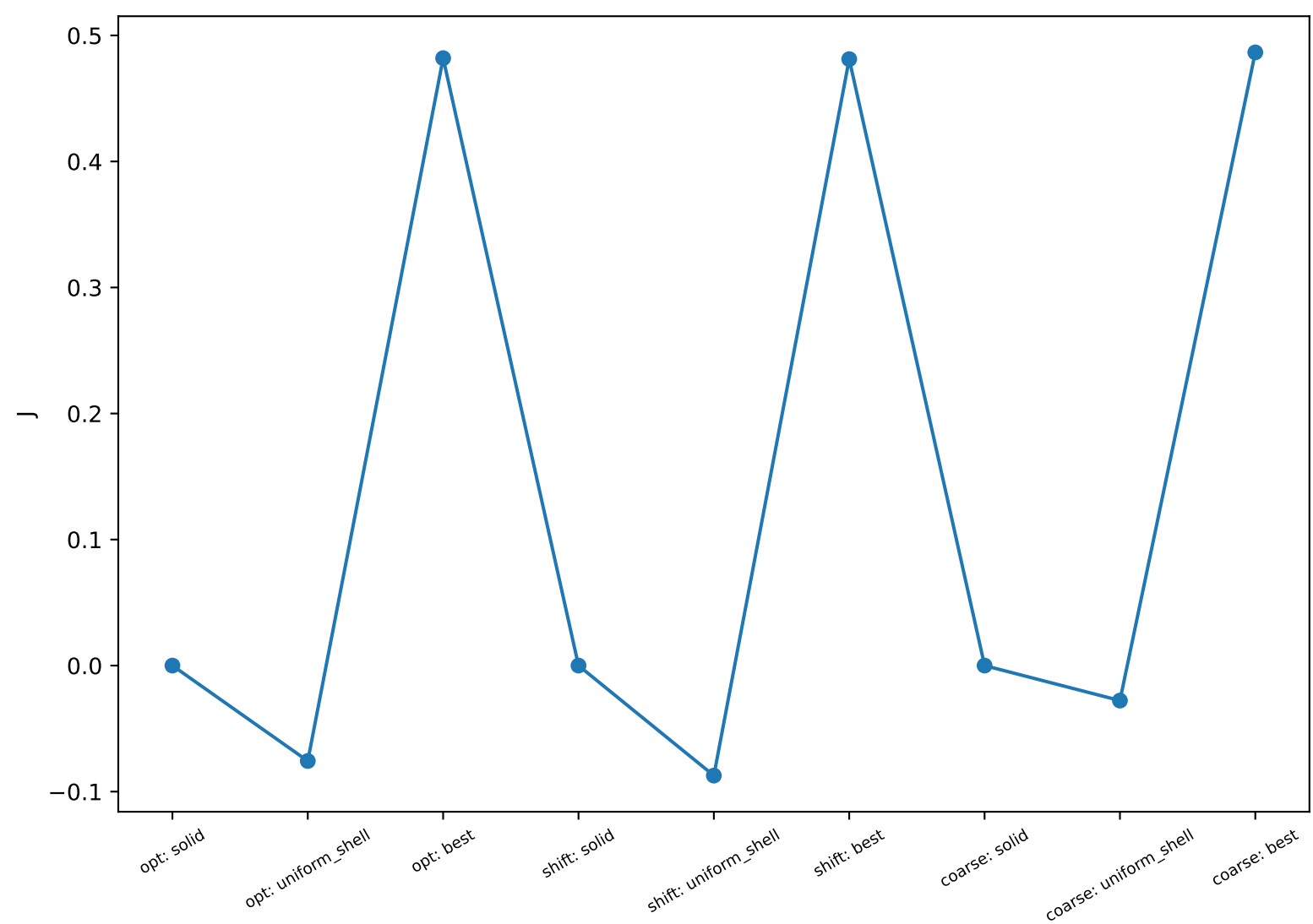
missing input: buckling mode shapes (no run-dir file stores eigenvectors)

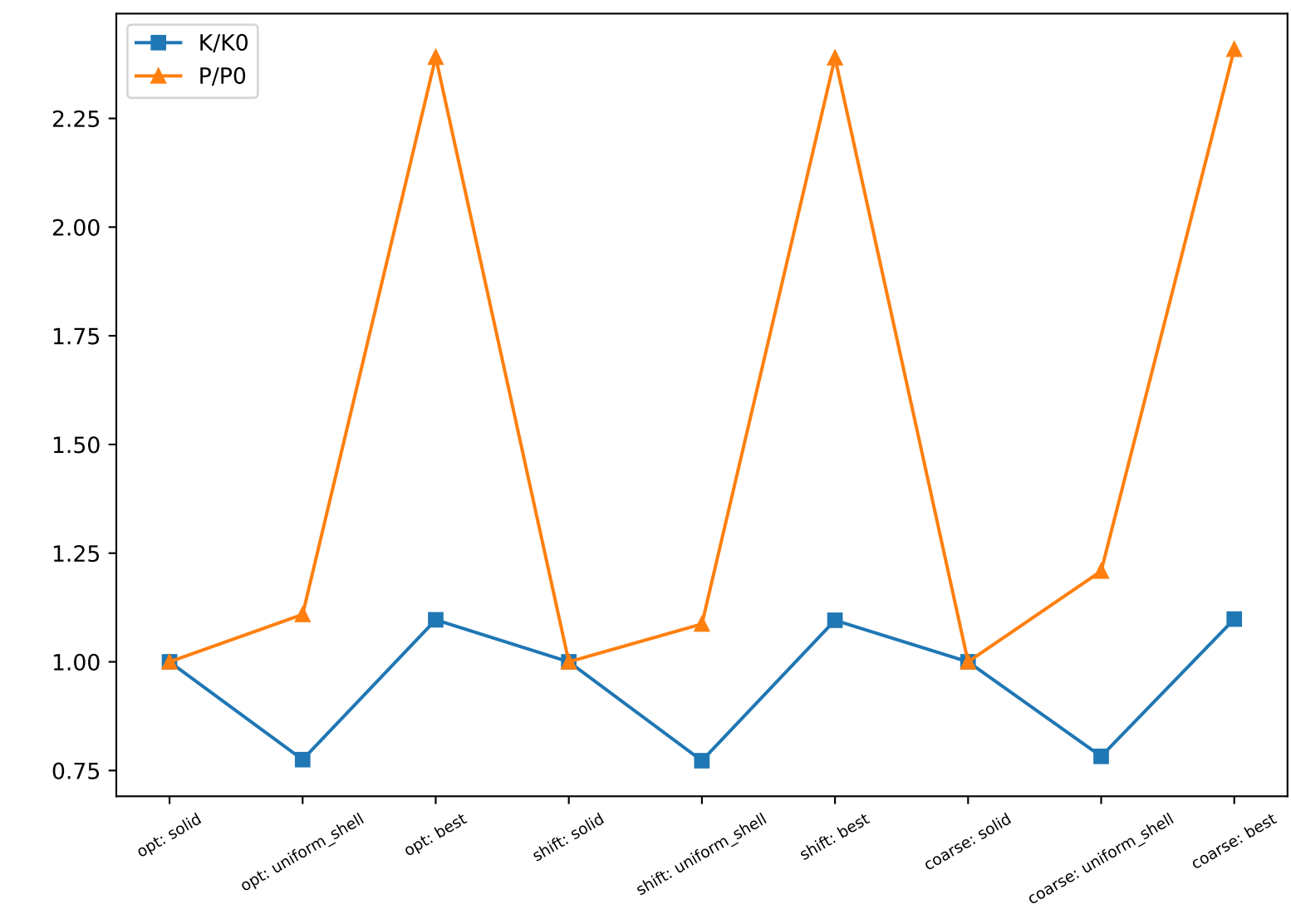

W_bar on free surface (Lambda/2 shown if in summary.json)
Lambda/2
W_bar
w [mm]
0.00
0.01
0.02
0.03
0.04
0.05
0.06
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5


W_bar
0.00
0.01
0.02
0.03
0.04
0.05
0.06


capacity per load case
14
12
10
8
6
4
2
0
lam_s
lam_b
softmin
1 (solid)
1 (cshell)


decision re-scored (no new FE: lam_b ~ kappa_b, lam_s ~ sigma_f)
20.0
10.0
0.0
-10.0
-20.0
sigma_f change [%]
0.6
0.7
0.85
1.0
kappa_b
0.48
0.46
0.44
0.42
0.40
0.38
0.36
J_star


J
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
opt: solid
opt: uniform_shell
opt: best
shift: solid
shift: uniform_shell
shift: best
coarse: solid
coarse: uniform_shell
coarse: best


K/K0
P/P0
2.25
2.00
1.75
1.50
1.25
1.00
0.75
opt: solid
opt: uniform_shell
opt: best
shift: solid
shift: uniform_shell
shift: best
coarse: solid
coarse: uniform_shell
coarse: best

12_validation: validation

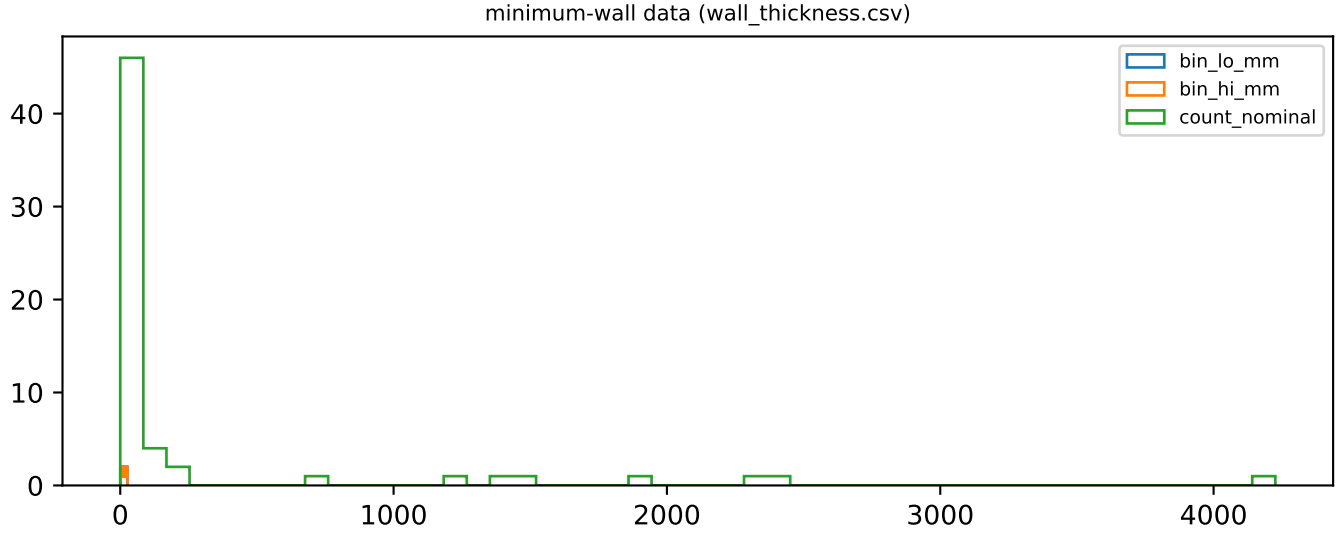
not available

missing input: validation.csv

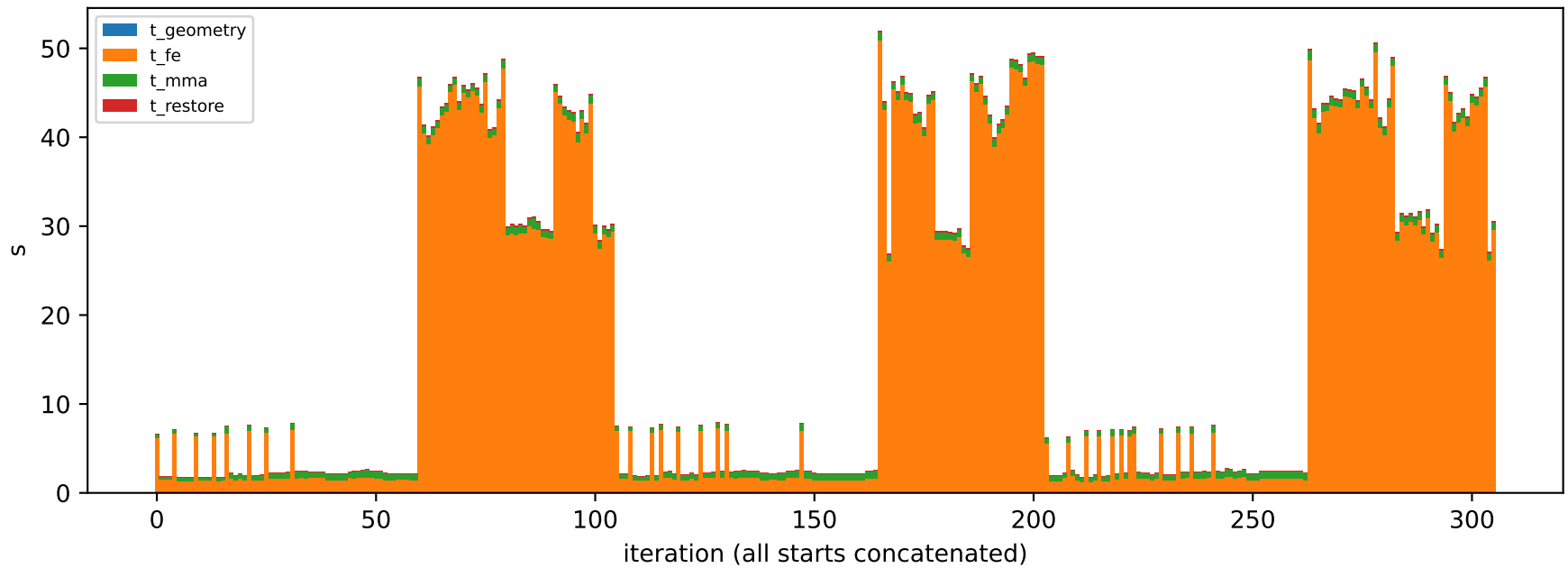
Voids and escape holes

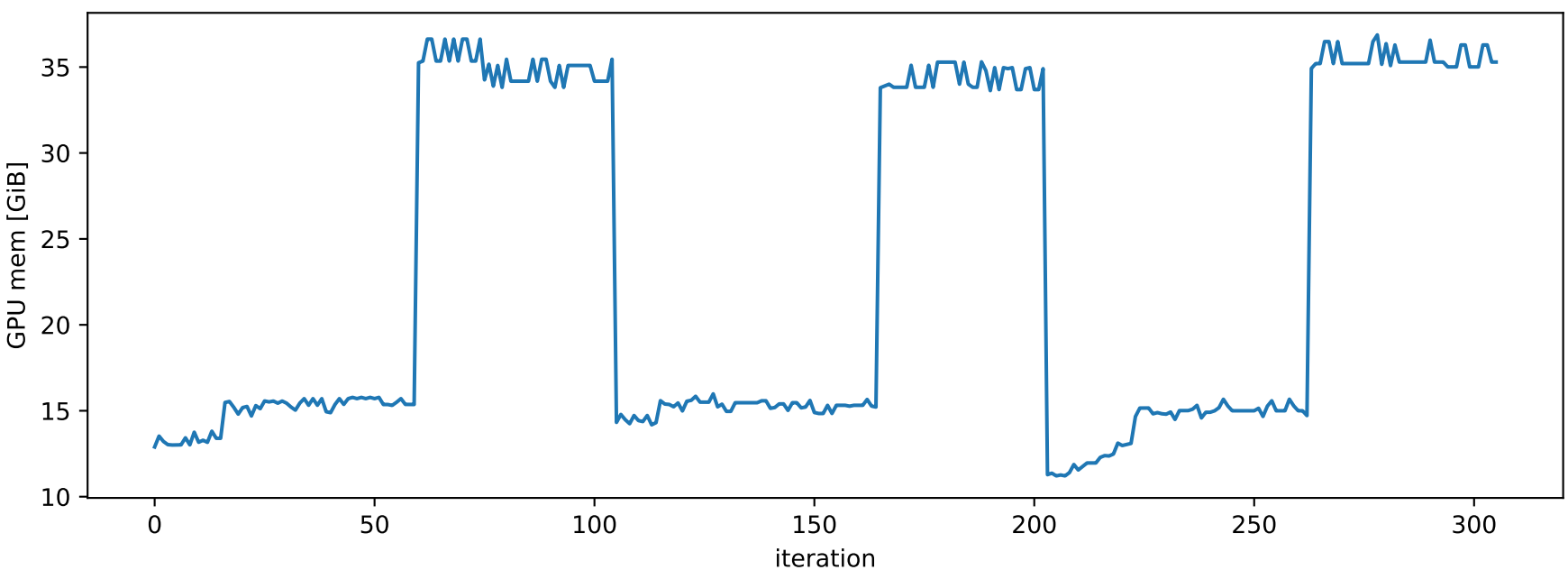
escape holes: 4

void_id	hole	x	y	z	nx	ny	nz	wall_mm	score	score_kind	void_volum
0	0	-13.48	50.63	43.98	-0.04685	0.9269	-0.3723	3.209	1.035	stress	2.712e+04
0	1	-12.47	-50.5	44.86	-0.1371	-0.9557	-0.2604	4.087	1.19	stress	2.712e+04
1	0	12.84	50.47	43.83	-0.02304	0.8542	-0.5194	3.084	1.297	stress	2.711e+04
1	1	77.87	-50.15	18.94	-0.04681	-0.9974	-0.05543	1.894	1.314	stress	2.711e+04



starts ['neutral', 'uniform', 'vshell']; summed iteration time 1.56 h; peak mem 36.9 GiB; GPU NVIDIA GH200 120GB; GPU-hours 2.540952054593298; jobs 7860610,7860611,7860612





Provenance

```
git commit  4af699e9192c0270c171164a2f3e797d3e15442c   dirty: True   (state when this report was built)
report built on host nid011075 at 2026-10-31T09:42:44+00:00; CUDA driver 13000, runtime 12090
run hosts / jobs (run_info.json): nid011284 job 7860610; nid010421 job 7860611; nid010187 job 7860612
STEP: {'path': '/home/b6bl/owenpeckham.b6bl/shell-lightweighting/.claude/worktrees/cshell-stage2/examples/step/Solid MBB.step', 'sha256': 'd0d7e5ba2ed69b9e8c8c94c15c00a3acbdd8d1c4ef56709aea1389561358fb82'}
config_used.json: {'path': '/home/b6bl/owenpeckham.b6bl/shell-lightweighting/.claude/worktrees/cshell-stage2/runs_cshell/FULL/MBB/config_used.json', 'sha256': 'f3e46395d4b0436f4f6fad39be8d8b6115a12df8b24f037d433ba3fc812d7455'}

source SHA-256:
  cshell/__init__.py                          a780b0223749228ac62b9efdc83804d1ea3f4e2fe6eea14df10b4283747da058
  cshell/band.py                              23f6ac86935f361c244b42f00ea0af1076c1d0b048ea67e52068b04f1fe31a56
  cshell/cli.py                               e9beb15df1b93eb07a463573ebd0c9cccdec6ddd6cebbf463cd69903159874a6
  cshell/config.py                            ac008f99c91c0c907415626bcfae25b58514b74acaed6114fd8266b3c633506a
  cshell/design.py                            e5c1d9b2db94efcd52d91cf71f87f67e7482280c9944d12d99c3b8a414e544de
  cshell/export.py                            376c4034bdc3b3407514fd6bba6b09d8ce7308aa5a1b18d9425476b57422ed33
  cshell/fe_sparse.py                         f9dc856bfb3c8289b6374a8ff0cd67300ac01f47704cf506385ced997ce22d31
  cshell/fea.py                               912a08b4e3c784bf2e039041c20bd7c3e524a338856722c264762782ce1895b8
  cshell/mma.py                               99e1516fcba4c60b808e273f4d276341ffd7a563014b7f7083b706e27f74ed35
  cshell/objective.py                         0bf352ab43fc12a4a111632382062e171baf290434eed416ce6667628ed14290
  cshell/optimise.py                          1393d4fd8f114d627a138887530db7562752394530 1525acef3028cac1ba3f53
  cshell/report.py                            765816dd322088475ec03eaa983af43c995d5eec983438f6aae6b70d4e62af51
  cshell/surface.py                           2e92a15030dea5893d16096740135a3af3203e2c03cc8e9b4a1c4459aa56d5ab
  cshell/validate.py                          2df7be08722ddfe4838c5d0bee0802659dd8e32391df11a9bdcb3c72dd21ee26
  cshell/collect.py                           aba3ff46fb6e7fb595a13098fad24bd1b9b264c2a7c8185816b93fb1950e2050
  continuous/reference/analysis.py            954c5feefc9a2bf9f190e26e2118b43d384988f03b8b177256ceb6ab1a6d7df5
  continuous/reference/gpu_analysis.py        04ce066539fa9396f3974efbafb8b66852f0ed73442353dd7ec9bd50e7456a39

packages:
  numpy               2.4.6
  scipy               1.17.1
  scikit-image        0.26.0
  skan                0.13.1
  libigl              2.6.3
  pyamg               5.3.0
  trimesh             5.2.0
  matplotlib          3.11.2
  build123d           0.13.0
  cupy                14.2.0
  pyvista             0.49.1
```